


*Robotics 2*

# Linear parametrization and identification of robot dynamics

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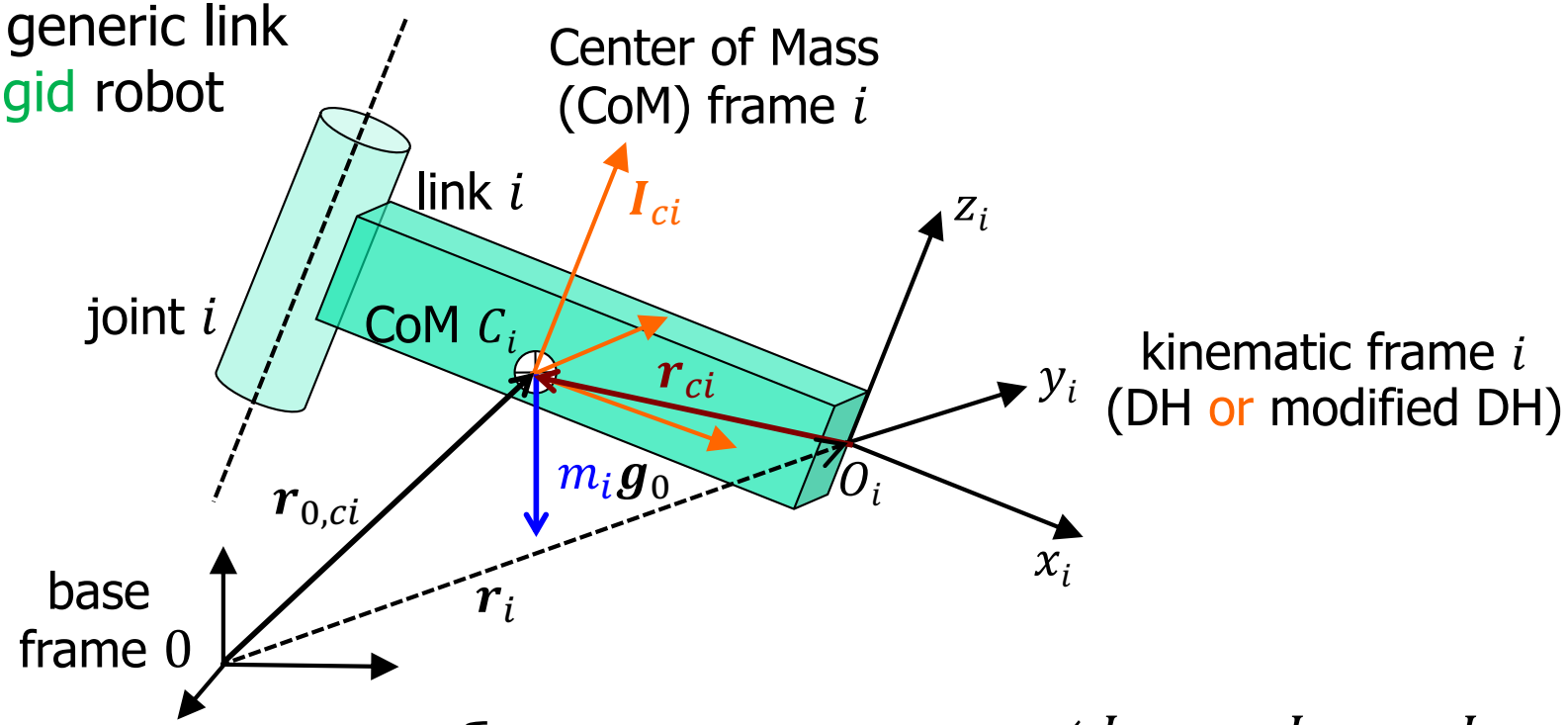
DIPARTIMENTO DI INGEGNERIA INFORMATICA
AUTOMATICA E GESTIONALE ANTONIO RUBERTI




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# Dynamic parameters of robot links

- consider a generic link of a fully rigid robot




- each link is characterized by 10 dynamic parameters

$$m_i \quad \boldsymbol{r}_{ci} = \begin{pmatrix} r_{xi} \\ r_{yi} \\ r_{zi} \end{pmatrix} \quad \boldsymbol{I}_{ci} = \begin{pmatrix} I_{ci,xx} & I_{ci,xy} & I_{ci,xz} \\ & I_{ci,yy} & I_{ci,yz} \\ \text{symm} & & I_{ci,zz} \end{pmatrix}$$

- however, robot dynamics depends only on some of these parameters and possibly in a nonlinear way (e.g., via the combination $I_{ci,zz} + m_i r_{zi}^2$)

# Dynamic parameters of robots




- both the kinetic energy and the gravity potential energy can be rewritten so that a new set of dynamic parameters appears only in a linear way
  - need to re-express link inertia and position of CoM in (any) known kinematic frame attached to the link (same orientation as the barycentric frame)
- fundamental kinematic relation

$$v_{ci} = v_i + \omega_i \times r_{ci} = v_i + S(\omega_i)\, r_{ci} = v_i - S(r_{ci})\, \omega_i$$

- kinetic energy of link $i$

$$T_i = \frac{1}{2} m_i v_{ci}^T v_{ci} + \frac{1}{2} \omega_i^T I_{ci} \omega_i \quad \Leftrightarrow \text{"reversing" König theorem now ...}$$

$$= \frac{1}{2} m_i (v_i - S(r_{ci})\omega_i)^T (v_i - S(r_{ci})\omega_i) + \frac{1}{2} \omega_i^T I_{ci} \omega_i$$

$$= \frac{1}{2} m_i v_i^T v_i + \frac{1}{2} \omega_i^T \underbrace{\left(I_{ci} + m_i S^T(r_{ci}) S(r_{ci})\right)}_{\text{Steiner theorem}} \omega_i - v_i^T S(m_i r_{ci}) \omega_i$$

$$I_i = \begin{pmatrix} I_{i,xx} & I_{i,xy} & I_{i,xz} \\ & I_{i,yy} & I_{i,yz} \\ \text{symm} & & I_{i,zz} \end{pmatrix}$$

# Standard dynamic parameters of robots




- gravitational potential energy of link $i$

$$U_i = -m_i g_0^T r_{0,ci} = -m_i g_0^T (r_i + r_{ci}) = -m_i g_0^T r_i - g_0^T (m_i r_{ci})$$

- by expressing vectors and matrices in frame $i$, both $T_i$ and $U_i$ will be linear in the set of 10 (constant) standard parameters $\boldsymbol{\pi}_i \in \mathbb{R}^{10}$

$$T_i = \frac{1}{2} m_i \, {}^i v_i^T \, {}^i v_i + m_i \, {}^i r_{ci}^T \, S({}^i v_i) \, {}^i \omega_i + \frac{1}{2} {}^i \omega_i^T \, {}^i I_i \, {}^i \omega_i$$

$$U_i = -m_i g_0^T r_i - g_0^T \, {}^0 R_i \, (m_i \, {}^i r_{ci})$$

mass of link $i$ (0-th order moment)

mass × CoM position of link $i$ (1-st order moment)

inertia of link $i$ (2-nd order moment)

$$\boldsymbol{\pi}_i = \begin{pmatrix} m_i \\ m_i \, {}^i r_{ci} \\ vect\{{}^i I_i\} \end{pmatrix} = (m_i \quad m_i {}^i r_{ci,x} \quad m_i {}^i r_{ci,y} \quad m_i {}^i r_{ci,z} \quad {}^i I_{i,xx} \quad {}^i I_{i,xy} \quad {}^i I_{i,xz} \quad {}^i I_{i,yy} \quad {}^i I_{i,yz} \quad {}^i I_{i,zz})^T$$

- since the E-L equations involve only linear operations on $T$ and $U$, also the robot dynamic model is linear in the standard parameters $\boldsymbol{\pi} \in \mathbb{R}^{10N}$

# Additional dynamic parameters




- beside the $n_\ell = 10$ parameters for each of the $n$ links of the robot

$$(m_i \quad m_i {}^i r_{ci,x} \quad m_i {}^i r_{ci,y} \quad m_i {}^i r_{ci,z} \quad {}^i I_{i,xx} \quad {}^i I_{i,xy} \quad {}^i I_{i,xz} \quad {}^i I_{i,yy} \quad {}^i I_{i,yz} \quad {}^i I_{i,zz})^T$$

  one can include in the identification problem

  - one inertia parameter for each of the $n$ motors, as reflected through its reduction ratio

$$M_{mi} = I_{mi} n_{ri}^2$$

  - $n_f$ dissipative parameters for each of the $n$ joints/transmissions, using viscous, Coulomb, and static friction models (and possibly more)

$$(B_{vi} \quad B_{ci} \quad B_{,i})^T \qquad (n_f = 3)$$

  - the above terms appear already in a linear form in the robot dynamic model

- the total number $p$ of dynamic parameters is

$$\Rightarrow \quad \boldsymbol{\pi}_i \in \mathbb{R}^{n_p} \quad n_p = n_\ell + 1 + n_f \quad \Rightarrow \quad \boldsymbol{\pi} \in \mathbb{R}^p \quad \boxed{p = n \cdot n_p}$$

# Linearity in the dynamic parameters




- using a $n \times p$ regression matrix $Y_\pi$ that depends only on kinematic quantities, the robot dynamic equations can always be rewritten linearly in the standard (and additional) dynamic parameters as

$$M(q)\ddot{q} + c(q,\dot{q}) + g(q) = Y_\pi(q,\dot{q},\ddot{q})\,\pi = u$$

$$\pi^T = (\pi_1^T \quad \pi_2^T \quad \cdots \quad \pi_n^T)$$

- the open kinematic chain structure of the manipulator implies that the $i$-th dynamic equation can depend only on the dynamic parameters of links from $i$ to $n \Rightarrow Y_\pi$ has a block upper triangular structure

$$Y_\pi(q,\dot{q},\ddot{q}) = \begin{pmatrix} Y_{11} & Y_{12} & \cdots & Y_{1n} \\ 0 & Y_{22} & \cdots & Y_{2n} \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & 0 & Y_{nn} \end{pmatrix}$$

with row vectors $Y_{ij}$ of size $1 \times n_p$

property: element $m_{ij}$ of $M(q)$ depends at most on $(q_{k+1}, \cdots, q_n)$, with $k = \min\{i,j\}$, and at most on the dynamic parameters of links $h$ to $n$, with $h = \max\{i,j\}$

# Linearity in the dynamic coefficients




- many standard parameters do not appear ("play no role") in the dynamic model of a given robot ⇒ the corresponding columns of $Y_\pi$ are 0 and are simply deleted
- some of the remaining $p_r$ $(< p)$ parameters appear only in fixed combinations with others ⇒ the corresponding columns of $Y_\pi$ are linearly dependent
- one can isolate $r \ll p_r$ combinations of parameters $\pi$ (corresponding to a set of $r$ structurally independent columns $Y_{\text{ind}}$ of $Y_\pi$) and partition matrix $Y_\pi$ in two blocks, the second containing dependent columns written as $Y_{\text{dep}} = Y_{\text{ind}}\Delta$, for a suitable $r \times (p_r - r)$ constant matrix $\Delta$

$$Y_\pi(q,\dot{q},\ddot{q})\,\boldsymbol{\pi} = \begin{pmatrix} Y_{\text{ind}} & Y_{\text{dep}} \end{pmatrix} \begin{pmatrix} \pi_{\text{ind}} \\ \pi_{\text{dep}} \end{pmatrix} = \begin{pmatrix} Y_{\text{ind}} & Y_{\text{ind}}\,\Delta \end{pmatrix} \begin{pmatrix} \pi_{\text{ind}} \\ \pi_{\text{dep}} \end{pmatrix}$$

$$= Y_{\text{ind}}\left(\pi_{\text{ind}} + \Delta\,\pi_{\text{dep}}\right) = \boxed{Y(q,\dot{q},\ddot{q})\,\boldsymbol{\rho}}$$

- the new parameters $\boldsymbol{\rho} = \pi_{\text{dep}} + \Delta\,\pi_{\text{dep}} \in \mathbb{R}^r$ are the dynamic coefficients of the robot dynamics, "the only that matter" —also called base parameters (W. Khalil)
- the minimal number $r$ of dynamic coefficients that are strictly needed can also be checked numerically (see → identification)

# Linear parametrization of robot dynamics




it is always possible to rewrite the dynamic model in the form

regression matrix

$\rho$ = vector of dynamic coefficients

$$M(q)\ddot{q} + c(q,\dot{q}) + g(q) = Y(q,\dot{q},\ddot{q})\,\rho = u$$

$n \times r$ $\qquad$ $r \times 1$

e.g., the heuristic grouping (found by inspection) on the planar 2R robot

see slides #12-14 in block 04_LagrangianDynamics_2

$$\begin{pmatrix} \ddot{q}_1 & c_2(2\ddot{q}_1 + \ddot{q}_2) - s_2(\dot{q}_2^2 + 2\dot{q}_1\dot{q}_2) & \ddot{q}_2 & c_1 & c_{12} \\ 0 & c_2\ddot{q}_1 + s_2\dot{q}_1^2 & \ddot{q}_1 + \ddot{q}_2 & 0 & c_{12} \end{pmatrix} \begin{pmatrix} \rho_1 \\ \rho_2 \\ \rho_3 \\ \rho_4 \\ \rho_5 \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

$$\rho_1 = I_{c1,zz} + m_1 d_{c1}^2 + I_{c2,zz} + m_2 d_{c2}^2 + m_2 l_1^2$$
$$\rho_2 = m_2 l_1 d_{c2}$$
$$\rho_3 = I_{c2,zz} + m_2 d_{c2}^2$$
$$\rho_4 = g_0(m_1 d_{c1} + m_2 l_1)$$
$$\rho_5 = g_0 m_2 d_{c2}$$

**Note:** 4 more coefficients $F_{V,i}$ and $F_{C,i}$ of viscous and Coulomb friction are added when including dissipative effects at the joints ("decoupled" terms, appearing only in $i$–th equation, for $i = 1,2$)

# Linear parametrization
## of a 2R planar robot ($n = 2$)




- being the kinematics known (i.e., $l_1$ and $g_0$), the number of dynamic coefficients is reduced since we can merge two coefficients into one

  $\rho_2 = m_2 l_1 d_{c2}$ & $\rho_5 = g_0 m_2 d_{c2}$ $\Rightarrow$ $\rho_2 = m_2 d_{c2}$ (factoring out $l_1$ and $g_0$)

- therefore, after regrouping, $r = 4$ dynamic coefficients are sufficient

$$\begin{pmatrix} \ddot{q}_1 & \underbrace{\begin{matrix} l_1 c_2(2\ddot{q}_1 + \ddot{q}_2) - l_1 s_2(\dot{q}_2^2 + 2\dot{q}_1\dot{q}_2) + g_0 c_{12} \\ l_1(c_2\ddot{q}_1 + s_2\dot{q}_1^2) + g_0 c_{12} \end{matrix}}_{l_1 Y_{2,\text{old}} + g_0 Y_{5,\text{old}}} & \ddot{q}_2 & \underbrace{\begin{matrix} g_0 c_1 \\ 0 \end{matrix}}_{g_0 Y_{4,\text{old}}} \\ 0 & & \ddot{q}_1 + \ddot{q}_2 & \end{pmatrix} \begin{pmatrix} \rho_1 \\ \rho_2 \\ \rho_3 \\ \rho_4 \end{pmatrix} = Y\rho = u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

$$\rho_1 = I_{c1,zz} + m_1 d_{c1}^2 + I_{c2,zz} + m_2 d_{c2}^2 + m_2 l_1^2 \qquad \rho_3 = I_{c2,zz} + m_2 d_{c2}^2$$

$$\rho_2 = m_2 d_{c2} \ (\ldots) l_1 \qquad \rho_4 = m_1 d_{c1} + m_2 l_1 \ (\ldots) g_0$$

- this (minimal) linear parametrization of robot dynamics is not unique, both in terms of the chosen set of dynamic coefficients $\rho$ and for the corresponding regression matrix $Y(q, \dot{q}, \ddot{q})$
- a systematic procedure for its derivation would be preferable

# Linear parametrization
## of a 2R planar robot ($n = 2$)




- in alternative to the previous heuristic method, apply the general procedure
  - neglect the presence of motors and of dissipative effects
  - for this robot, $p = 10n = 20$ standard parameters $\boldsymbol{\pi}$ are defined for the two links
  - assuming that the CoMs are on the kinematic axis of the links, only $p_r = 5$ such parameters appear explicitly (having set ${}^{i}r_{ci,x} = -l_i + d_{ci}$)

$$\boxed{Y_\pi \, \pi = u} \qquad \pi = \begin{pmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \\ \pi_4 \\ \pi_5 \end{pmatrix} = \begin{pmatrix} m_1 d_{c1} \\ I_{c1,zz} + m_1 d_{c1}^2 \\ m_2 \\ m_2 d_{c2} \\ I_{c2,zz} + m_2 d_2^2 \end{pmatrix} = \begin{pmatrix} m_1 d_{c1} \\ I_{1,zz} \\ m_2 \\ m_2 d_{c2} \\ I_{2,zz} \end{pmatrix} \begin{matrix} \} \text{ link 1} \\ \\ \} \text{ link 2} \end{matrix}$$

check the differences with $Y$ and $\pi$ of slide #8 (up to reordering)

$$Y_\pi = \begin{pmatrix} g_0 c_1 & \ddot{q}_1 & l_1^2 \ddot{q}_1 + l_1 g_0 c_1 & l_1 c_2 (2\ddot{q}_1 + \ddot{q}_2) - l_1 s_2 (2\dot{q}_1 \dot{q}_2 + \dot{q}_2^2) + g_0 c_{12} & \ddot{q}_1 + \ddot{q}_2 \\ 0 & 0 & 0 & l_1 (c_2 \ddot{q}_1 + s_2 \dot{q}_1^2) + g_0 c_{12} & \ddot{q}_1 + \ddot{q}_2 \end{pmatrix}$$

- the third column of $Y_\pi$ (associated to $\pi_3 = m_2$) is not structurally independent

$$Y_{\pi 3} = l_1 \, Y_{\pi 1} + l_1^2 \, Y_{\pi 2} \quad \Rightarrow \quad \begin{aligned} Y_{\text{ind}} &= (Y_{\pi 1} \;\; Y_{\pi 2} \;\; Y_{\pi 4} \;\; Y_{\pi 5}) \\ Y_{\text{dep}} &= Y_{\pi 3} \\ Y_{\text{dep}} &= Y_{\text{ind}} \, \Delta \end{aligned} \qquad \Delta = \begin{pmatrix} l_1 \\ l_1^2 \\ 0 \\ 0 \end{pmatrix} \quad \Rightarrow \quad \rho = \begin{pmatrix} \rho_1 \\ \rho_2 \\ \rho_3 \\ \rho_4 \end{pmatrix} = \underbrace{\begin{pmatrix} \pi_1 \\ \pi_2 \\ \pi_4 \\ \pi_5 \end{pmatrix}}_{\pi_{\text{ind}}} + \Delta \, \underbrace{\pi_3}_{\pi_{\text{dep}}}$$

# Linear parametrization
## of a 2R planar robot ($n = 2$)




- the new linear parametrization of the robot dynamics in terms of $r = 4$ dynamic coefficients (again minimal!) is

$$\boxed{Y(q, \dot{q}, \ddot{q})\, \rho = u}$$

$$Y = \begin{pmatrix} g_0 c_1 & \ddot{q}_1 & l_1 c_2 (2\ddot{q}_1 + \ddot{q}_2) - l_1 s_2 (\dot{q}_2^2 + 2\dot{q}_1 \dot{q}_2) + g_0 c_{12} & \ddot{q}_1 + \ddot{q}_2 \\ 0 & 0 & l_1 (c_2 \ddot{q}_1 + s_2 \dot{q}_1^2) + g_0 c_{12} & \ddot{q}_1 + \ddot{q}_2 \end{pmatrix} = Y_{\text{ind}}$$

$$\rho = \begin{pmatrix} \rho_1 \\ \rho_2 \\ \rho_3 \\ \rho_4 \end{pmatrix} = \begin{pmatrix} \pi_1 + l_1 \pi_3 \\ \pi_2 + l_1^2 \pi_3 \\ \pi_4 \\ \pi_5 \end{pmatrix} = \begin{pmatrix} m_1 d_{c1} + m_2 l_1 \\ I_{c1,zz} + m_1 d_{c1}^2 + m_2 l_1^2 \\ m_2 d_{c2} \\ I_{c2,zz} + m_2 d_2^2 \end{pmatrix} = \begin{pmatrix} m_1 d_{c1} + m_2 l_1 \\ I_{1,zz} + m_2 l_1^2 \\ m_2 d_{c2} \\ I_{2,zz} \end{pmatrix} \in \mathbb{R}^4$$

compare with $Y$ and $\rho$ of slide #9
(which is also minimal but different)

- determining a minimal parameterization (i.e., minimizing $r$) is important for
  - experimental identification of dynamic coefficients
  - adaptive/robust control design in the presence of uncertain parameters

# Identification of dynamic coefficients




- to "use" the model, one needs to know the numeric values of all robot dynamic coefficients
  - robot manufacturers provide at most only a few principal dynamic parameters (e.g., link masses)
- estimates can be found with CAD tools (e.g., assuming uniform mass)
- friction coefficients are (slowly) varying over time
  - lubrication of joints/transmissions
- for an added payload (attached to the E-E)
  - a change in the 10 standard dynamic parameters of last link …
  - … implies a variation of (almost) all robot dynamic coefficients
- preliminary identification experiments are needed
  - robot in motion (dynamic issues, not just static or geometric ones!)
  - only the robot dynamic coefficients can be identified (and not all link standard parameters!)

# Identification experiments




1. choose a motion trajectory $q_d(t)$ that is sufficiently "exciting", i.e.,
   - explores the robot workspace and involves all components in the robot dynamic model
   - is periodic, with multiple frequency components
2. execute this motion (approximately) by means of a control law
   - taking advantage of any available information on the robot model
   - often $u = K_P(q_d - q) + K_D(\dot{q}_d - \dot{q})$ (PD, no model information used)
3. measure $q$ (encoders) in $n_c$ time instants (and, if available, also $\dot{q}$)
   - joint velocity $\dot{q}$ and acceleration $\ddot{q}$ can be estimated later offline by numerical differentiation (use of non-causal filters is admissible)
4. with such measures/estimates, evaluate the regression matrix $Y$ (on the left) and use the applied commands $u$ (on the right) in the expression

$$Y\big(q(t_k), \dot{q}(t_k), \ddot{q}(t_k)\big)\,\rho = u(t_k) \quad k = 1, \cdots, n_c$$

# Least Squares (LS) identification

- set up the system of linear equations

$$n_c \times n \left\{ \begin{pmatrix} Y\left(q(t_1), \dot{q}(t_1), \ddot{q}(t_1)\right) \\ \vdots \\ Y\left(q(t_{n_c}), \dot{q}(t_{n_c}), \ddot{q}(t_{n_c})\right) \end{pmatrix} \right. \rho = \begin{pmatrix} u(t_1) \\ \vdots \\ u(t_{n_c}) \end{pmatrix} \quad \Leftrightarrow \quad \bar{Y}\rho = \bar{u}$$

- sufficiently "exciting" trajectories, large enough number of samples ($n_c \times n \gg r$), and their suitable selection/position guarantee that rank $\bar{Y} = r$ (full column rank)
- solution by pseudoinversion of matrix $\bar{Y}$

$$\rho = \bar{Y}^{\#}\bar{u} = (\bar{Y}^T\bar{Y})^{-1}\bar{Y}^T\bar{u} \quad (\in \mathbb{R}^r)$$

- one can also use a weighted pseudoinverse, to consider the different levels of noise in the collected measures

# Additional remarks on LS identification

- it is convenient to preserve the block (upper) triangular structure of the regression matrix, by "stacking" all time evaluations in row-by-row sequence of the original $Y$ matrix

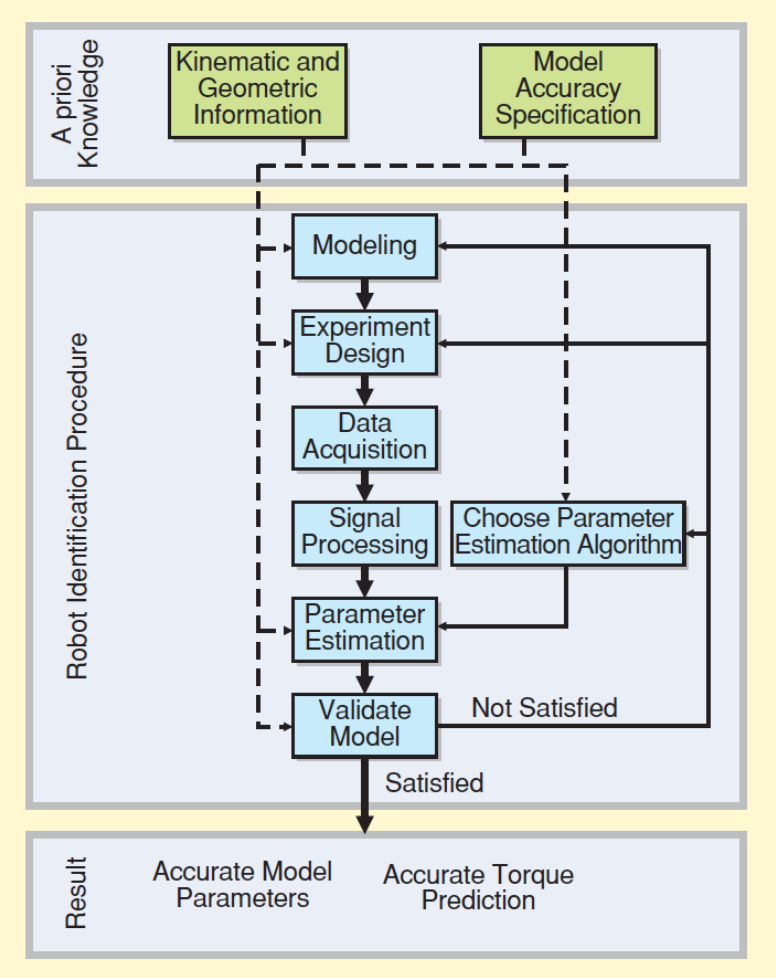
$$n \times \begin{cases} n_c \\ \\ \\ \\ \\ n_c \end{cases} \begin{pmatrix} Y_1(q(t_1), \dot{q}(t_1), \ddot{q}(t_1)) \\ \vdots \\ Y_1\left(q(t_{n_c}), \dot{q}(t_{n_c}), \ddot{q}(t_{n_c})\right) \\ Y_2(q(t_1), \dot{q}(t_1), \ddot{q}(t_1)) \\ \vdots \\ Y_2\left(q(t_{n_c}), \dot{q}(t_{n_c}), \ddot{q}(t_{n_c})\right) \\ \vdots \\ Y_N(q(t_1), \dot{q}(t_1), \ddot{q}(t_1)) \\ \vdots \\ Y_N\left(q(t_{n_c}), \dot{q}(t_{n_c}), \ddot{q}(t_{n_c})\right) \end{pmatrix} \rho = \begin{pmatrix} u_1(t_1) \\ \vdots \\ u_1(t_{n_c}) \\ u_2(t_1) \\ \vdots \\ u_2(t_{n_c}) \\ \vdots \\ u_N(t_1) \\ \vdots \\ u_N(t_{n_c}) \end{pmatrix} \Leftrightarrow \boxed{\bar{Y}\rho = \bar{u}}$$

$$\bar{Y} = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \begin{matrix} \\ \\ \\ \\ n_c \\ n_c \end{matrix}$$

- numerical check of full column rank is more robust ⇔ rank = $r$ (# of columns)

- further practical hints
  - outlier data can be eliminated in advance (when building $\bar{Y}$)
  - if sufficiently rich friction models are not included in $Y\rho$, discard the data collected at joint velocities close to zero

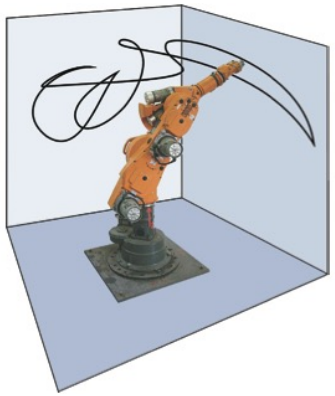
# Summary on dynamic identification

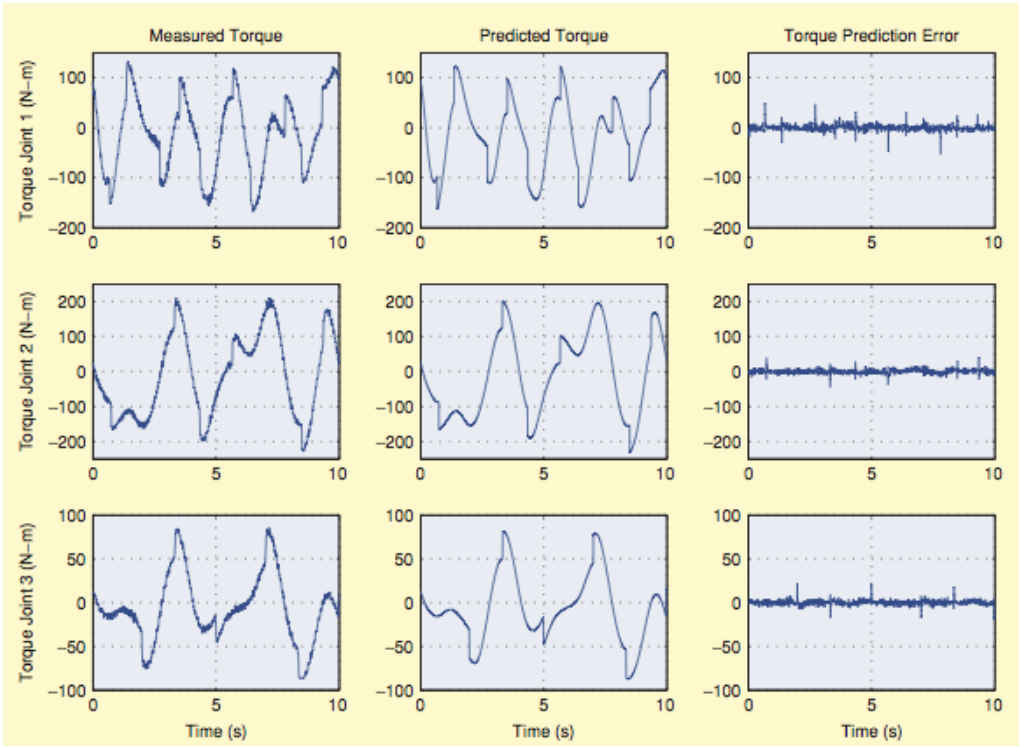



J. Swevers, W. Verdonck, and J. De Schutter:
"Dynamic model identification for industrial robots"
IEEE Control Systems Mag., Oct 2007

KUKA IR 361 robot and optimal excitation trajectory







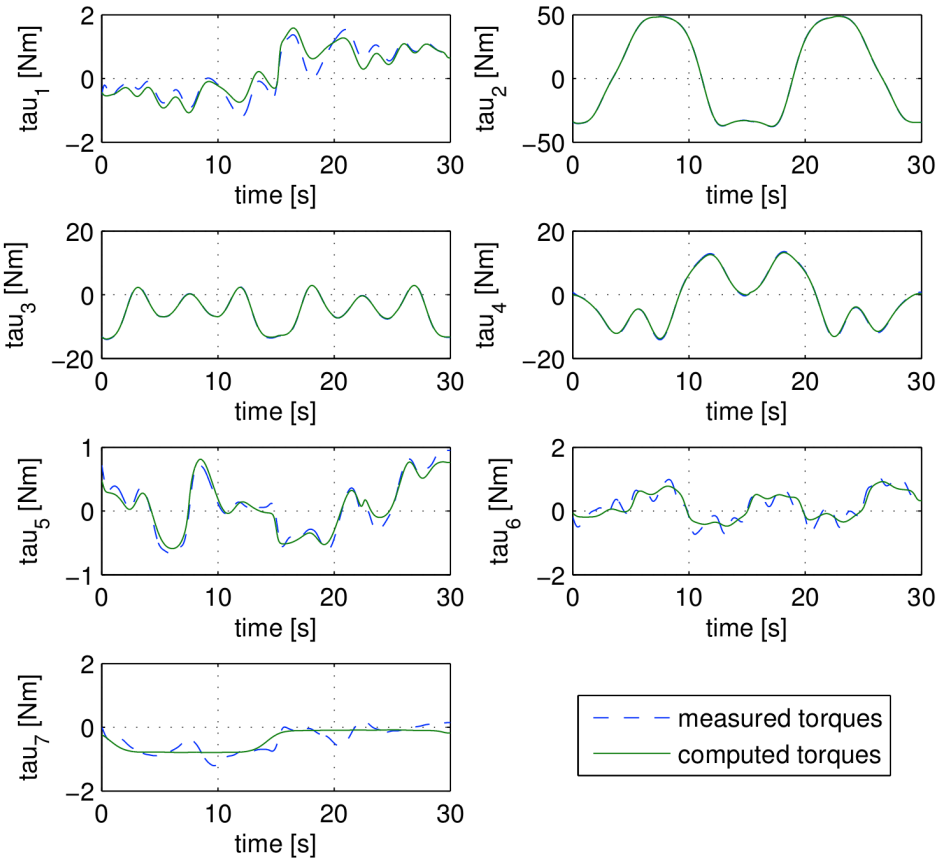

results after identification (first three joints only)

# Dynamic identification of KUKA LWR4

video




data acquisition for identification




dynamic coefficients: 30 inertial, 12 for gravity

C. Gaz, F. Flacco, A. De Luca:
"Identifying the dynamic model used by the KUKA LWR:
A reverse engineering approach", IEEE ICRA 2014

validation after identification (for all 7 joints):
on new desired trajectories, compare
torques computed with the identified model
and torques measured by joint torque sensors

# Identification of LWR4 gravity terms

using the linear parametrization, gravity terms can also be identified separately

$$\boldsymbol{\rho}_g = \begin{pmatrix} c_{7y}m_7 \\ c_{7x}m_7 \\ c_{6x}m_6 \\ c_{6z}m_6 + c_{7z}m_7 \\ c_{5z}m_5 - c_{6y}m_6 \\ c_{5x}m_5 \\ c_{5y}m_5 + c_{4z}m_4 + d_2(m_5 + m_6 + m_7) \\ c_{4x}m_4 \\ c_{4y}m_4 + c_{3z}m_3 \\ c_{2x}m_2 \\ c_{3x}m_3 \\ c_{2z}m_2 - c_{3y}m_3 + d_1(m_3 + m_4 + m_5 + m_6 + m_7) \end{pmatrix}$$

position of CoM of link $i$

$\mathbf{c_i} = (c_{ix}\ c_{iy}\ c_{iz})^T$

two DH parameters $d_1,\ d_2$

$$\boldsymbol{g(q)} = \boldsymbol{Y}_g(\boldsymbol{q})\boldsymbol{\rho}_g$$

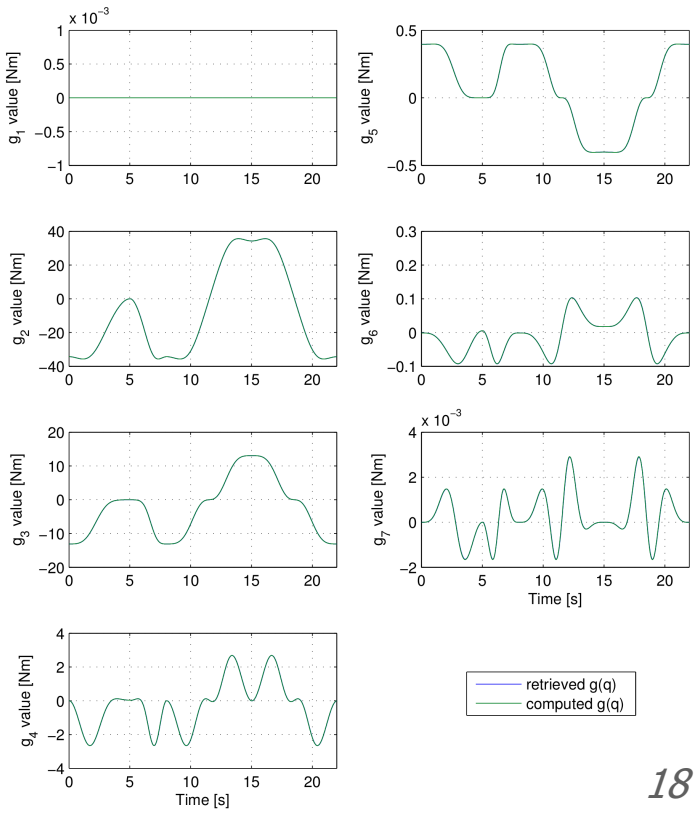
symbolic expressions of gravity-related dynamic coefficients

some small values may also be discarded …

$$\hat{\boldsymbol{\rho}}_g = \begin{pmatrix} 9.5457 \times 10^{-4} \\ -2.9826 \times 10^{-4} \\ 8.3524 \times 10^{-4} \\ 0.0286 \\ -0.0407 \\ -6.5637 \times 10^{-4} \\ 1.334 \\ -0.0035 \\ -4.7258 \times 10^{-4} \\ 0.0014 \\ 9.4532 \times 10^{-4} \\ 3.4568 \end{pmatrix}$$

numerical values identified through experiments

gravity joint torques prediction/evaluation on new validation trajectory (in practice, identical!)

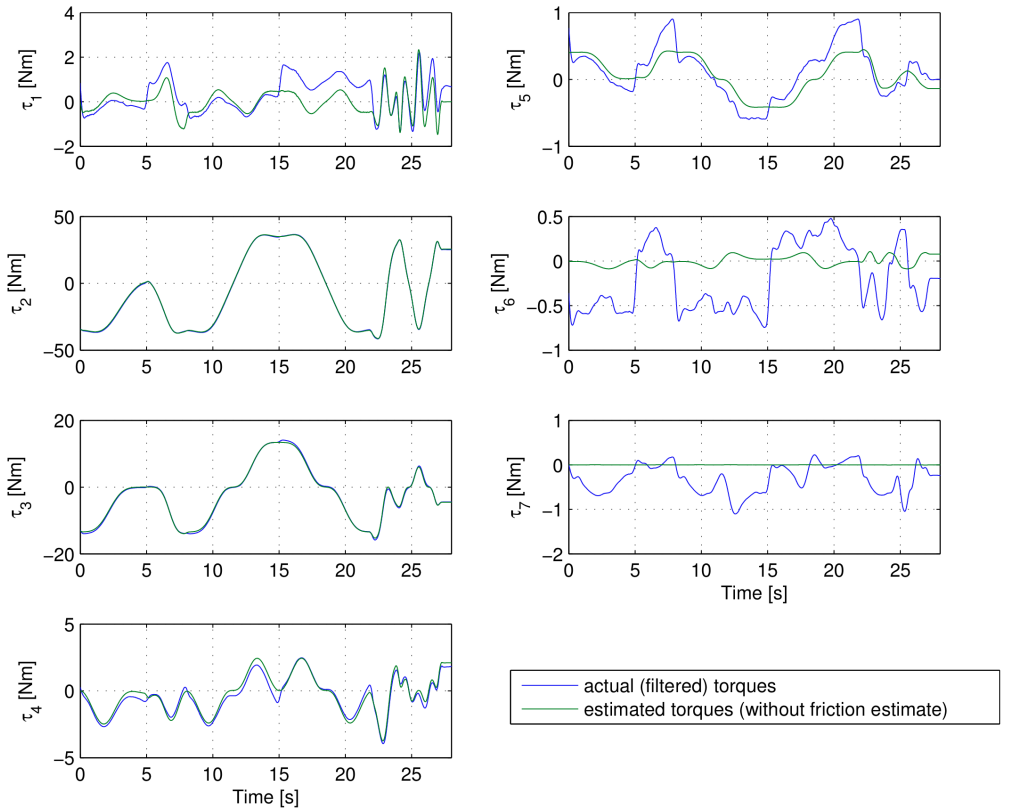


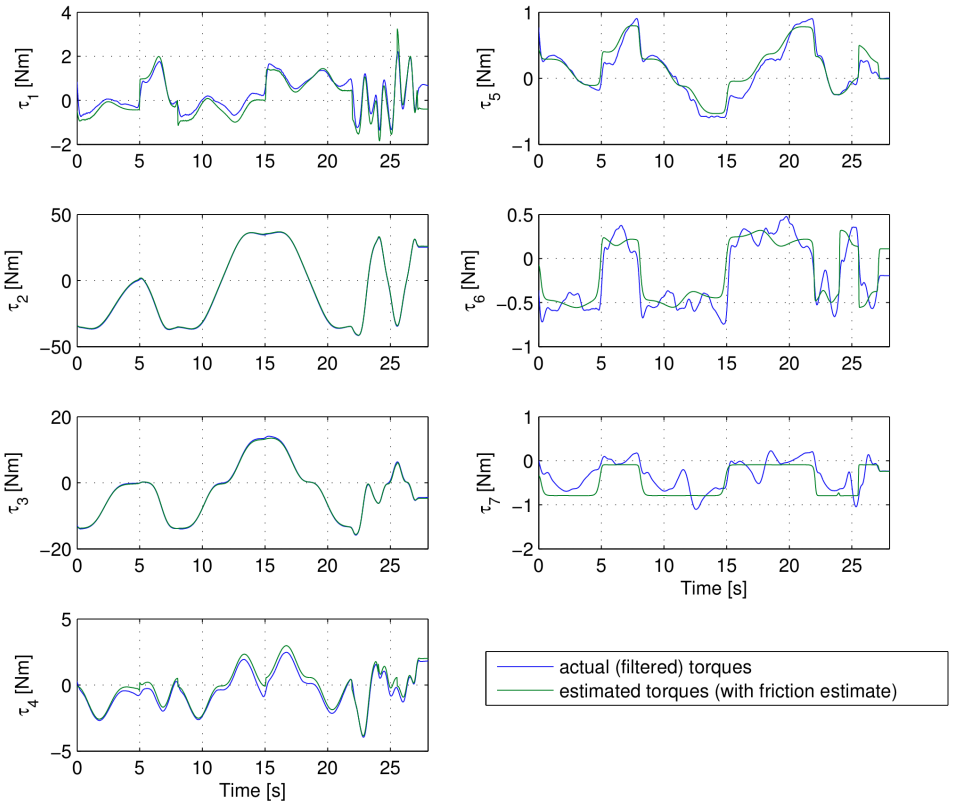
# Role of friction in identification

KUKA LWR4 dynamic model estimation vs. joint torque sensor measurement

$$M(q)\ddot{q} + c(q,\dot{q}) + g(q) = \tau - \tau_{friction}$$

$\tau_{meas}$



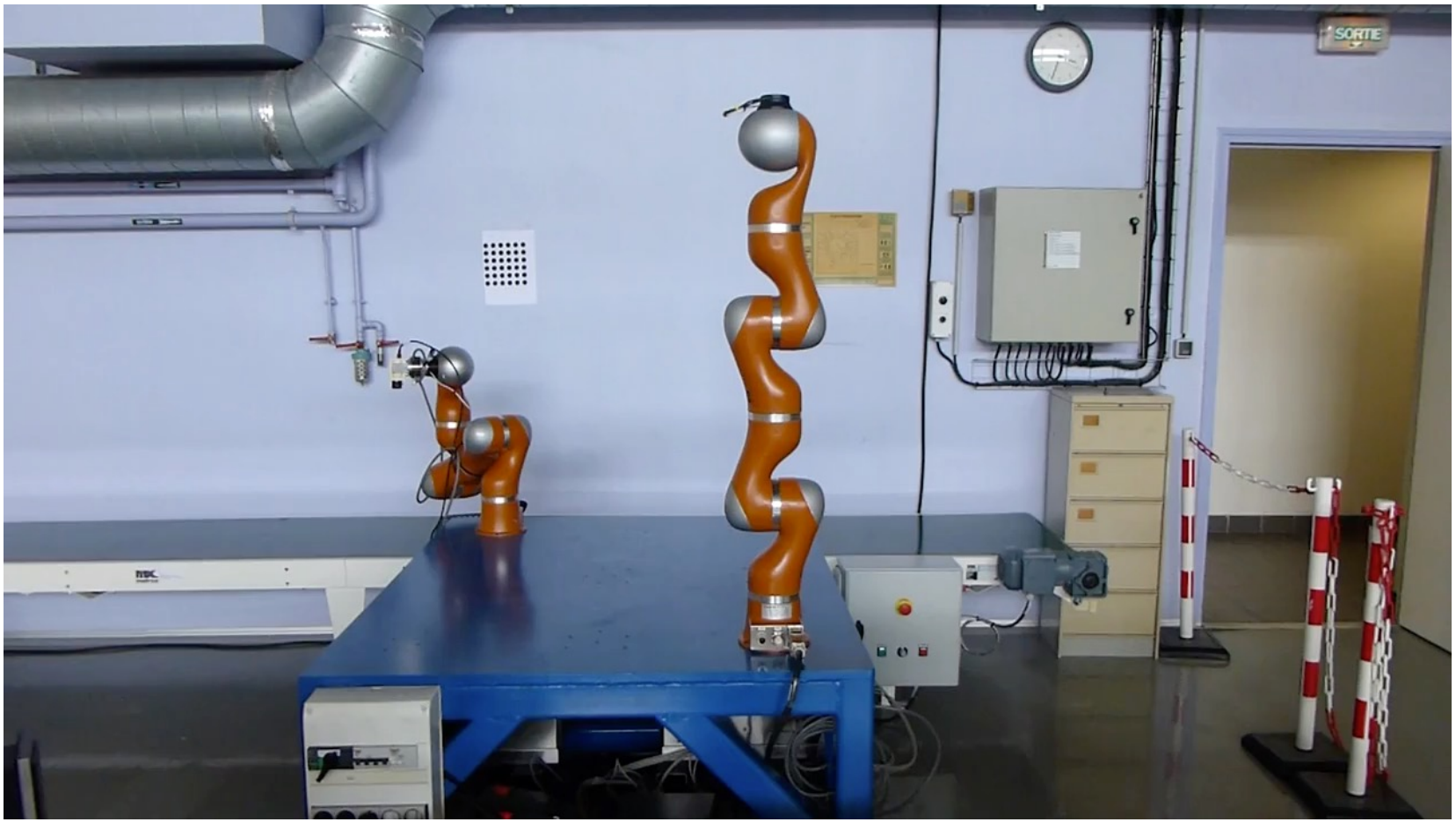




without the use of a joint friction model

including an identified joint friction model

with 3 parameters $\varphi_{1,j}$, $\varphi_{2,j}$, $\varphi_{3,j}$ appearing in a nonlinear way $\longrightarrow$ $\tau_{f,j}(\dot{q}_j) = \frac{\varphi_{1,j}}{1+e^{-\varphi_{2,j}(\dot{q}_j+\varphi_{3,j})}} - \frac{\varphi_{1,j}}{1+e^{-\varphi_{2,j}\varphi_{3,j}}}$

# Dynamic identification of KUKA LWR4




video

using more dynamic robot motions for model identification

J. Hollerbach, W. Khalil, M. Gautier: "Chap. 6: Model Identification", Springer Handbook of Robotics (2nd Ed), 2016
free access to multimedia extension: http://handbookofrobotics.org

# Adding a payload to the robot




- in several industrial applications, changes in the robot payload are often needed
  - using different tools for various technological operations such as polishing, welding, grinding, ...
  - pick-and-place tasks of objects having unknown mass
- what is the rule of change for dynamic parameters when there is an additional payload?
  - do we obtain again a linearly parameterized problem?
  - does this property rely on some specific choice of reference frames (e.g., conventional or modified DH)?

# Rule of change in dynamic parameters




- only the dynamic parameters of the link where a load is added will change (typically, added to the last one – link $n$ – as payload)
  - last link dynamic parameters: $m_n$ (mass), $\boldsymbol{c}_n = (c_{nx}\ c_{ny}\ c_{nz})^T$ (center of mass), $\boldsymbol{I}_n$ (inertia tensor expressed w.r.t. frame $n$)
  - payload dynamic parameters: $m_L$ (mass), $\boldsymbol{c}_L = (c_{Lx}\ c_{Ly}\ c_{Lz})^T$ (center of mass), $\boldsymbol{I}_L$ (inertia tensor expressed w.r.t. frame $n$)

- mass $\quad m_n \rightarrow m_n + m_L$

- center of mass $\quad c_{ni} m_n \rightarrow \dfrac{c_{ni} m_n + c_{Li} m_L}{m_n + m_L} (m_n + m_L) = c_{ni} m_n + c_{Li} m_L$

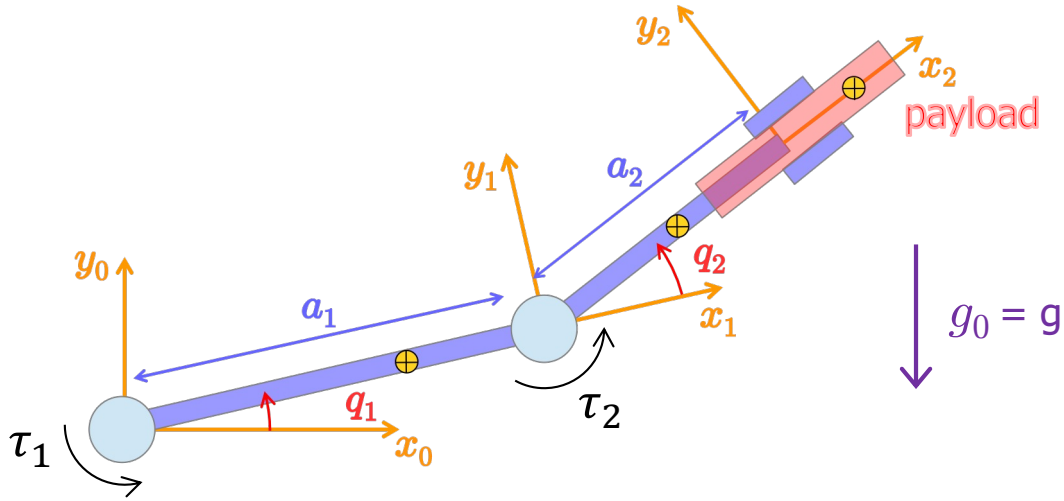
  (weighted average) $\qquad$ where $i = x,\ y,\ z$

- inertia tensor $\quad \boldsymbol{I}_n \rightarrow \boldsymbol{I}_n + \boldsymbol{I}_L$

  valid only if tensors are expressed w.r.t. the same reference frame (i.e., frame $n$)!

- linear parametrization is preserved with any kinematic convention (the parameters of the last link will always appear in the form shown above)

# Example: 2R planar robot with payload



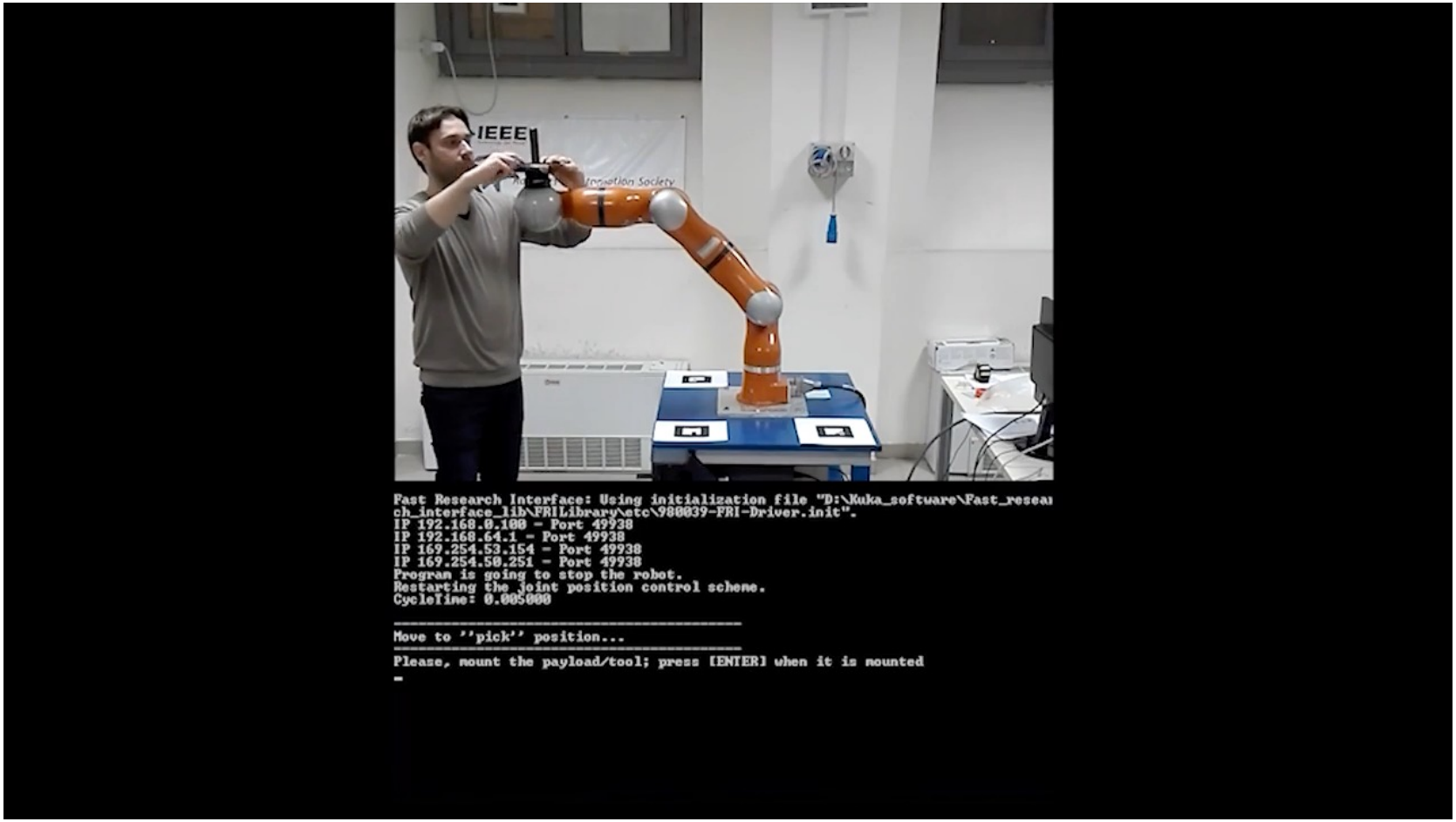

unloaded robot dynamics $\boldsymbol{Y\rho = \tau}$ loaded robot dynamics $\boldsymbol{Y^L\rho^L = \tau}$

$$\boldsymbol{\rho} = \begin{pmatrix} \frac{1}{2}\left(m_2 a_2^2 + I_{2zz}\right) + a_2 c_{2x} m_2 \\ c_{2x} m_2 + a_2 m_2 \\ c_{2y} m_2 \\ \frac{1}{2}\left(I_{1zz} + a_1^2 m_1 + a_1^2 m_2\right) + a_1 c_{1x} m_1 \\ c_{1x} m_1 + a_1 m_1 + a_1 m_2 \\ c_{1y} m_1 \end{pmatrix} \Rightarrow \boldsymbol{\rho}^L = \begin{pmatrix} \frac{1}{2}\left(a_2^2 (m_2 + m_L) + I_{2zz} + I_{Lzz}\right) + a_2 (c_{2x} m_2 + c_{Lx} m_L) \\ c_{2x} m_2 + c_{Lx} m_L + a_2 (m_2 + m_L) \\ c_{2y} m_2 + c_{Ly} m_L \\ \frac{1}{2}\left(I_{1zz} + a_1^2 m_1 + a_1^2 (m_2 + m_L)\right) + a_1 c_{1x} m_1 \\ c_{1x} m_1 + a_1 m_1 + a_1 (m_2 + m_L) \\ c_{1y} m_1 \end{pmatrix}$$

Note 1: position of the center of mass of the two links and of the payload can be asymmetric

Note 2: link inertia & center of mass are expressed in the DH kinematic frame attached to the link (e.g., $I_{2zz}$ is the inertia of the second link around the axis $z_2$)

# Validation on the KUKA LWR4 robot




video

C. Gaz, A. De Luca: "Payload estimation based on identified coefficients of robot dynamics – with an application to collision detection" IEEE IROS 2017, Vancouver, September 2017

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KUKA LWR4 (7R)





Universal Robot UR10 (6R)



Franka Emika Panda (7R)